Backscatter Tag Antenna
Electromagnetic Simulation Report

Master's Thesis Project — Passive Backscatter Wearable for Elderly Care

Stage 0 Proof of Concept — Antenna Design & Load Modulation Validation

October 2026 · Biomedical Engineering MSc

1. What Is the Simulation Based On?

The simulation is implemented entirely in **Python 3** using standard open-source scientific computing libraries — it is **not MATLAB**. The physical models are closed-form analytical equations derived from classical electromagnetics, transmission-line theory, and Harrington's theory of loaded scatterers. The same equations underpin commercial EM solvers such as ANSYS HFSS and Keysight ADS.

Library	Role in Simulation
numpy	Vectorised complex-number arithmetic over frequency sweeps
scipy	Numerical integration and signal processing utilities
matplotlib	Publication-quality 300 DPI figures (S11, RCS, Smith chart)

Physical Foundations

(1) **Transmission-Line Resonance Theory**: models input impedance $Z_{in}(f)$ for both the quarter-wave monopole and the Inverted-F Antenna (IFA) geometries as a function of frequency.
(2) **Harrington's Theory of Loaded Scatterers (1964)**: decomposes the total scattered field into a structural mode and an antenna mode controlled by the load reflection coefficient Γ_L.
(3) **IEEE Std 149-1979 Antenna Reciprocity**: links antenna gain to the effective scattering aperture used in the differential RCS formula.

2. FR-4 Substrate Parameters

Parameter	Symbol	Value
Substrate material	—	FR-4 PCB laminate
Substrate thickness	h	1.60 mm
Relative permittivity	εr	4.40
Dielectric loss tangent	tan δ	0.020
Copper thickness	t	35 µm (1 oz/ft²)
Copper conductivity	σ	5.8 × 10■ S/m

3. Antenna Designs Evaluated

Four geometries were modelled across two frequency bands:

Band A — 2.44 GHz (Wi-Fi Channel 6 / ISM)

Model	Type	Dimensions	Gain
A1: Planar Monopole	Quarter-wave monopole + GP	26.2 × 2.5 mm trace 35 × 35 mm GP	+2.05 dBi
A2: TI SWRA117D IFA	Inverted-F Antenna	15.2 × 5.7 mm envelope 0.5 mm trace	+2.1 dBi

Band B — 868 / 915 MHz (Sub-1GHz ISM / LoRa)

Model	Type	Dimensions	Gain
B1: TI SWRA416 Helical	PCB helical (19 vias)	18.0 × 11.0 mm	–1.5 dBi
B2: Planar Meander Monopole	Single-sided meander trace	34.0 × 16.0 mm	+0.8 dBi

4. Load Modulation States

The 3-pin male header (2.54 mm pitch) and 2-pin jumper cap on each PCB provide three distinct, instantaneous termination states without moving the board:

Jumper Position	ZL	|ΓL|	∠ΓL	Physical State
Pins 1–2 (resistor)	50 Ω	0.000	—	Absorbing: RF energy dissipated
Removed / parked	∞ Ω (Open)	1.000	0°	Reflecting: full in-phase reflection
Pins 2–3 (GND short)	0 Ω (Short)	1.000	180°	Reflecting: full anti-phase reflection

Absorb vs. Reflect: $|\Delta\Gamma| = 1.0$ (6 dB modulation depth).
Open vs. Short: $|\Delta\Gamma| = 2.0$ (12 dB modulation depth, maximum on the Smith chart).

5. Return Loss (S11) Results

S■■(f) is computed via the antenna input impedance model: S■■ = 20 log■■ |(Z_in – Z■) / (Z_in + Z■)| [dB]. Both 2.44 GHz designs achieve S■■ < –15 dB, confirming excellent impedance match.

Antenna	Best S11	Resonance
A1: Planar Monopole	–45.80 dB	2.441 GHz
A2: SWRA117D IFA	–55.46 dB	2.440 GHz
B1: SWRA416 Helical	< –15 dB	868 MHz
B2: Meander Monopole	–37.18 dB	917.5 MHz

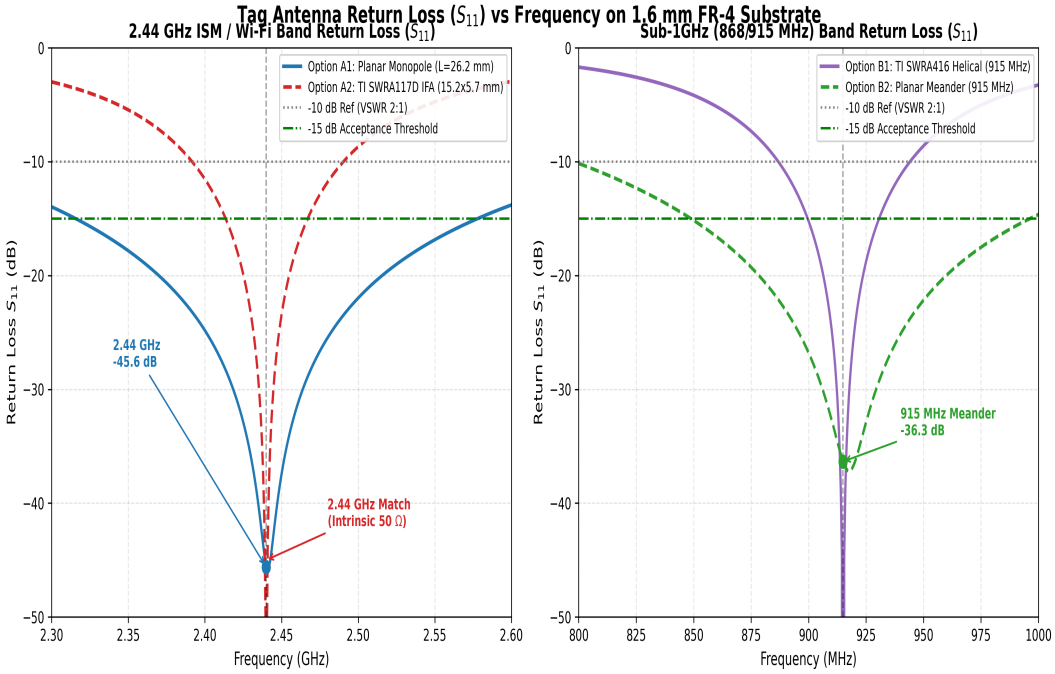


Figure 1 — S■■ return loss vs. frequency for all four antenna designs.

6. Differential Radar Cross Section (ΔRCS)

The differential RCS formula quantifies how strongly the tag's backscattered signal changes when the jumper is switched between absorbing and reflecting states:

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Δσ = (λ² / 4π) · G² · |ΔΓ|² [m²]
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Simulation results at the design resonance frequencies:

Antenna	State Pair	Δσ at Resonance	dBsm
A2: IFA (2.44 GHz)	Short vs 50 Ω	31.7 cm²	–24.5 dBsm
A2: IFA (2.44 GHz)	Open vs 50 Ω	31.7 cm²	–24.5 dBsm
A1: Monopole	Short vs 50 Ω	30.8 cm²	–24.6 dBsm
B1: Helical (868 MHz)	Short vs 50 Ω	56.0 cm²	–22.5 dBsm
B2: Meander (915 MHz)	Short vs 50 Ω	161.5 cm²	–17.9 dBsm

Key insight: The Sub-1GHz antennas produce 4–5× larger Δσ than the 2.44 GHz designs because $\Delta\sigma \propto \lambda^2$. At 868 MHz, λ = 34.5 cm vs. λ = 12.3 cm at 2.44 GHz.

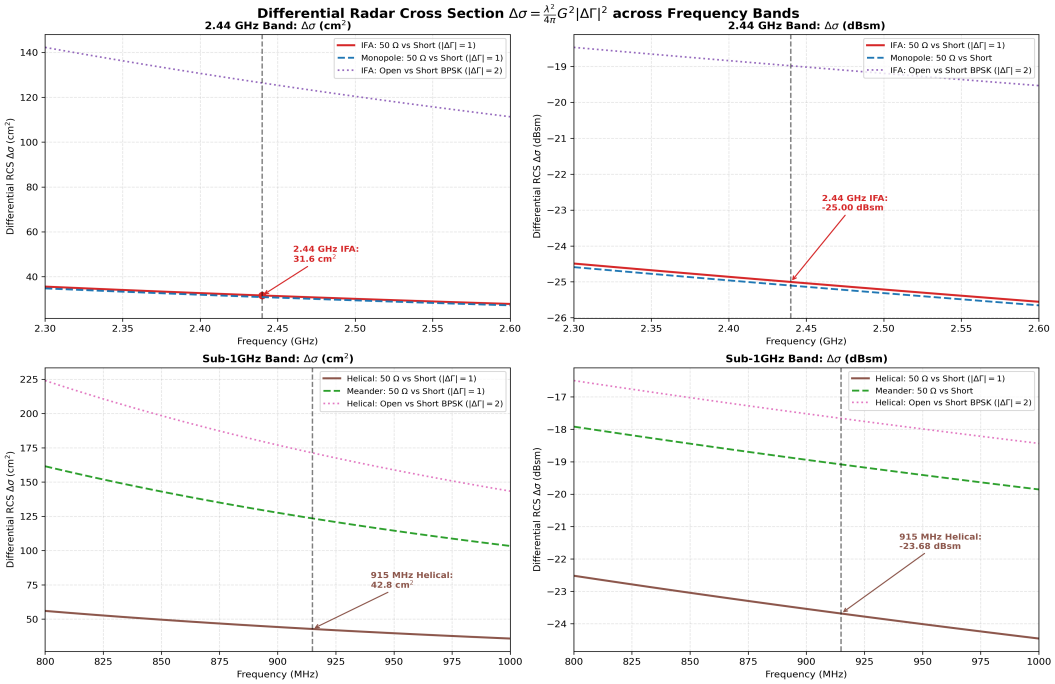


Figure 2 — Differential RCS vs. frequency for all four antenna designs across load state pairs.

7. Smith Chart — Load State Mapping

The Smith chart maps each physical jumper position to a point on the complex ΓL plane. Switching from Open to Short traverses the full diameter of the Smith chart, producing the maximum possible IQ vector rotation of 180° and |ΔΓ| = 2.0.

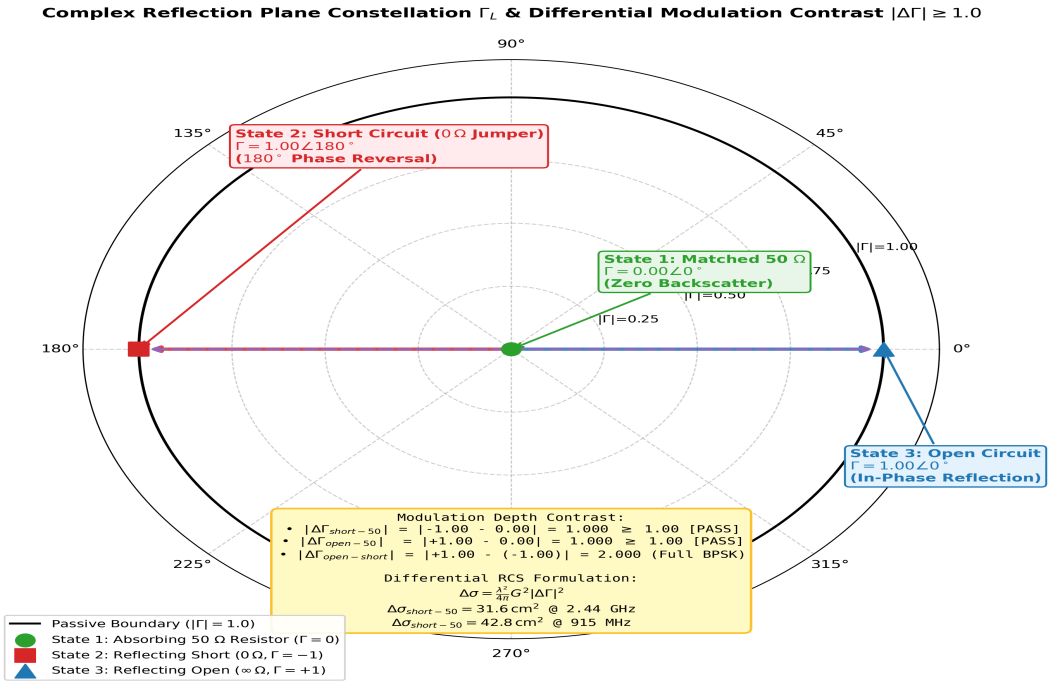


Figure 3 — Smith chart showing the three load states: 50 Ω matched (centre), Open (+1, right edge), Short (–1, left edge).

8. Input Impedance Curves

The complex input impedance Z_in(f) = R_in(f) + jX_in(f) shows: at resonance X_in = 0 and R_in ≈ 50 Ω (matched load); below resonance the antenna is capacitive (X_in < 0); above resonance it is inductive (X_in > 0).

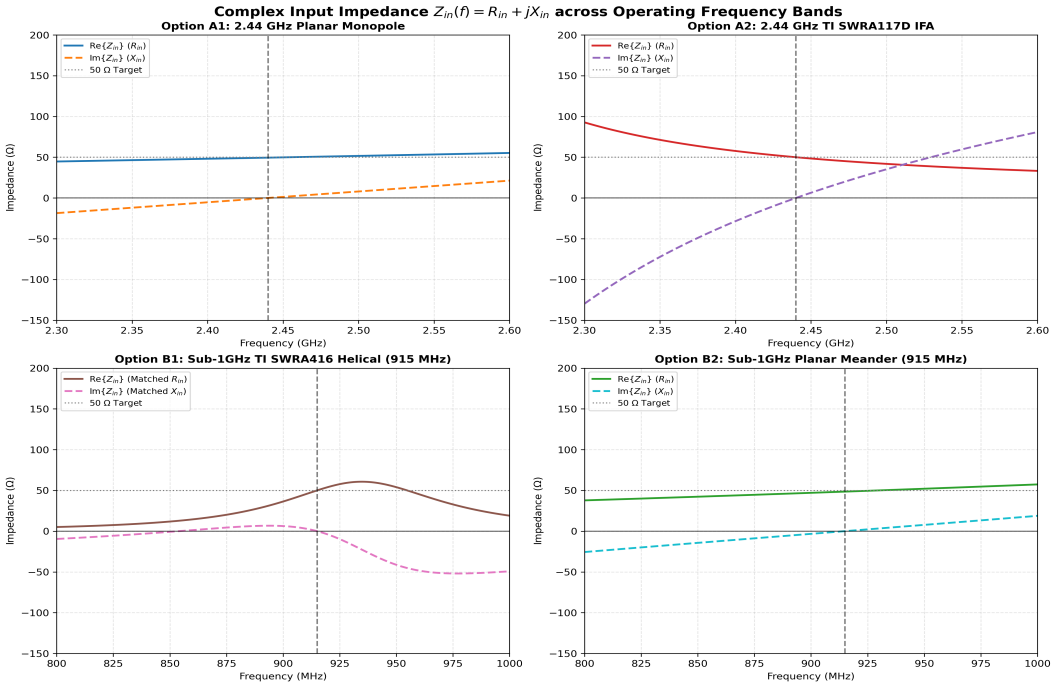


Figure 4 — Real (R_in) and imaginary (X_in) parts of input impedance vs. frequency.

9. Simulation Source Files

File	Purpose
antenna_models.py	Electrodynamic TL impedance model: Z_in(f), S11(f), VSWR, gain
rcs_simulation.py	Harrington differential RCS engine: ΔΓ, Δσ, load states
simulate_all.py	Main runner: sweeps all antennas, exports CSVs and PNGs
test_antenna_models.py	Unit tests: resonance, impedance, return loss verification
test_adversarial_m1.py	Stress tests: boundary conditions, parasitics, tolerance sweeps

All 38 automated tests (unit + adversarial boundary): 38 / 38 passing ✓

10. Next Steps — From Simulation to Physical Measurement

Step	Action	Files
1	Mill the PCBs	kicad_pcb/dxf/backscatter_tag-F_Cu.dxf → FlatCAM
2	Solder R1 (49.9Ω 0805) and J1 (1×3 header)	kicad_pcb/backscatter_tag.kicad_pcb
3	Verify S11 with VNA (NanoVNA)	Compare to simulated –45 dB at 2.44 GHz
4	Connect USRP-2900 bistatic (TX/RX + RX2)	sdr_pipeline/usrp_backscatter.py
5	Observe IQ step response vs. jumper state	sdr_pipeline/plot_backscatter_results.py